# Abiram R

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## EDUCATION

| Institution | Degree / Board | Passing Year | Score / CGPA |
|---|---|---|---|
| Indian Institute of Technology Hyderabad | B.Tech | 2028 | 7.05 |
| Vishwa Sishya Vidyodaya Senior Secondary School | Class XII (CBSE) | 2024 | 93.40% |
| Vishwa Sishya Vidyodaya Senior Secondary School | Class X (CBSE) | 2022 | 91.80% |

## SCHOLASTIC ACHIEVEMENTS

- Ranked in the **Top 20** in the **Optiver Trade-a-thon** with over **1k+ participants** across the country.

## PROJECTS

### HNSW - Face and Image Retrieval System

***Tech Stack:*** *C++20 | Python | FastAPI | SQLite/PostgreSQL | OpenCV | Pillow | InsightFace (dlib) | REST APIs | Git*

- **Built an end-to-end face retrieval system** for 1,000+ event photos, implementing **HNSW from scratch** to enable sub-linear approximate nearest-neighbor search over high-dimensional face embeddings.
- **Engineered a scalable ingestion pipeline** for bulk upload, thumbnail generation, multi-face detection, embedding extraction, persistent HNSW graph construction, and a **SQLite/PostgreSQL metadata layer** for efficient image resolution.
- **Developed FastAPI-based REST APIs** to retrieve nearest neighbors from the custom HNSW index, serving compressed thumbnails and metadata for responsive performance.
- **Optimized retrieval** using configurable HNSW parameters and **Recall@K benchmarking** against brute-force search, with real-time matching during photo ingestion.

### Adaptive ML-Based Cache Eviction System

*C++20 | Python (scikit-learn) | Multithreading | doctest*

- Built a **multithreaded C++ key-value store** supporting **SET/GET/DEL with TTL expiration**, using hash maps and lists for **1M+ keys**
- Replaced **LRU and LFU rules** with an **ML-based eviction policy** that predicts object reuse and ranks candidates by future access
- Created **Zipfian, uniform, temporal-locality, bursty, and non-stationary workloads** to test the policy across changing patterns
- Reimplemented **C++ inference** and **benchmarked against LRU/LFU** on cache hit rate, throughput, latency, and overhead

### Deep Learning-Based Image Restoration & Super-Resolution

*Python / PyTorch / OpenCV / NumPy / Matplotlib / scikit-image / LPIPS / DIV2K*

- Engineered a **PyTorch U-Net** with **residual blocks** and **skip connections** to restore **noisy, blurry, and low-resolution images**
- Built a **configurable degradation pipeline** combining **blur, downsampling, noise, and JPEG artifacts** to generate **paired training data**
- Benchmarked **SRCNN, residual CNN, and U-Net** against **bicubic interpolation** using **PSNR, SSIM, and LPIPS** metrics
- Evaluated restoration robustness through **ablations** on **loss functions, degradation severity, and scale factors** across **DIV2K images**

## RELEVANT COURSES

**CS:** Introduction to Programming, Artificial Intelligence, ML Fundamentals, Computer Vision for Chemical & Life Sciences

**Mathematics:** Calculus I & II, Elementary Linear Algebra, Differential Equations, Numerical Methods

**Engineering & Systems:** Thermodynamic Laws & Phase Transitions, Transport Phenomena, Fluid Mechanics

## SKILLS

**Languages:** C++, C, Python, SQL

**Frameworks/Libraries:** PyTorch, scikit-learn, FastAPI, OpenCV, NumPy, Pandas, Matplotlib, scikit-image

**Databases:** SQLite, PostgreSQL, MySQL

**Developer Tools:** Git, GitHub, Jupyter Notebook, Google Colab, Overleaf, doctest

**Creative:** Figma, Adobe Lightroom, Adobe Photoshop, DaVinci Resolve

## POSITIONS OF RESPONSIBILITY

**Outreach Head | Alumni Cell, IIT Hyderabad**

- Led outreach for **Legacy Meet**, bringing together **30+ accomplished alumni** for an **interactive session**.
- **Built and maintained alumni relationships**, driving **participation and successful engagements**.

**InFocus Overall Head | Cultural Council, IIT Hyderabad**

- Lead the photography club of the institute to a distinguished tenure conducting **10+ events** attracting over **300+ active participants**.
- Directed club operations and implemented structured initiatives, resulting in a **40% increase** in fresher engagement.
- Organized **NIRA**, a nationwide photography competition with **750+ participants**, with outreach across **40+ colleges**..

**Photography Cup Captain | IICM 8.0 Contingent, IIT Hyderabad**

- Coordinated team **logistics, schedules, and communication** with the organizing committee, ensuring seamless event execution.
- Achieved **4th** in Theme-Based Photography and **8th** in Photostory among **23 IITs**.

## EXTRACURRICULAR

- Led the **Tamil New Year 2026 organizing committee**, managing a **3 lakh budget** and coordinating **30+ sports and cultural events** with **foot fall over 600+** including students, faculty, and their families.
- **NSO Basketball player**, actively participating in institute training and matches.
- Regular participant in institute and local marathons and community fun runs.